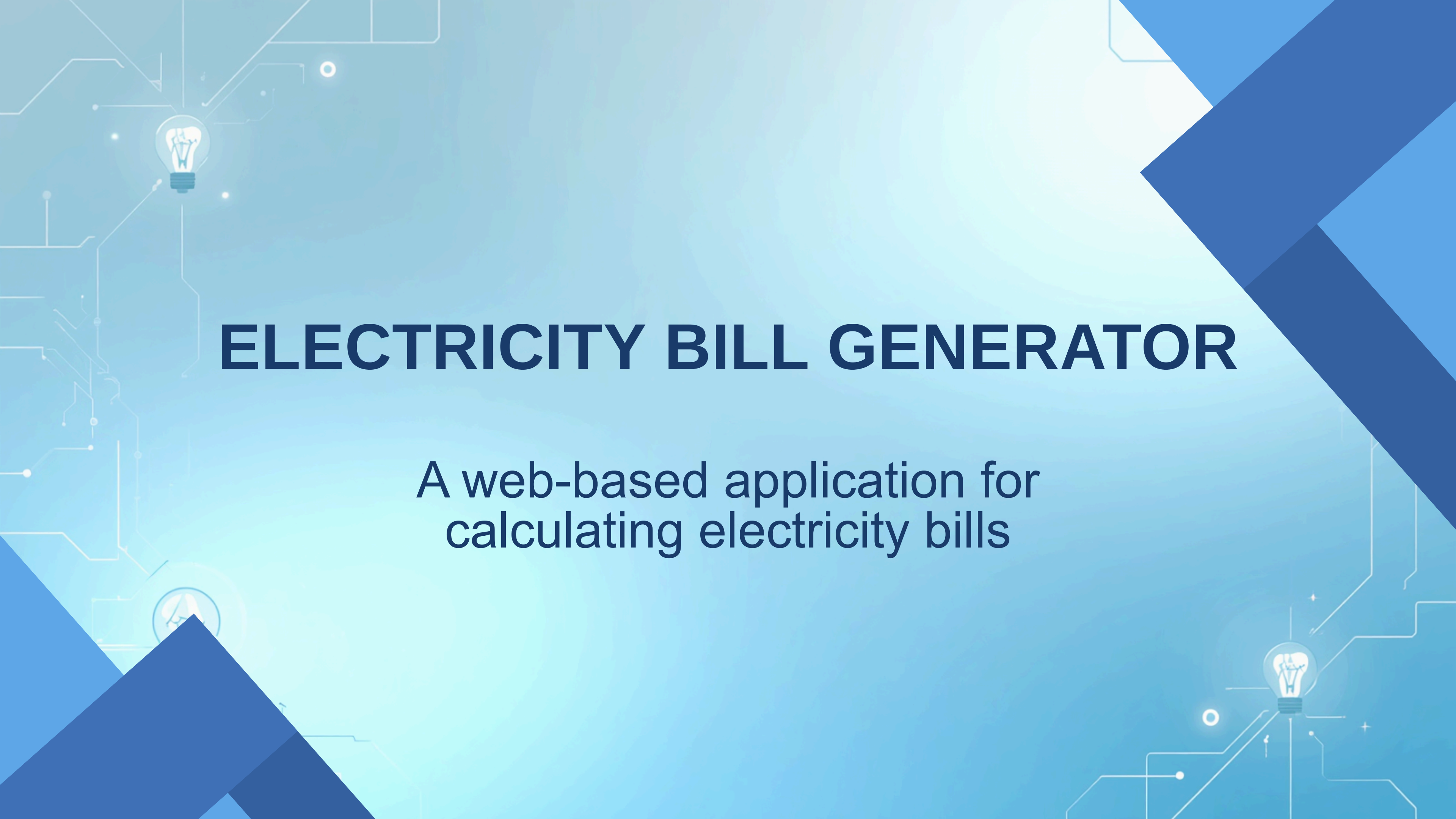


The background is a light blue gradient with faint white circuit lines. There are two glowing lightbulbs, one in the top left and one in the bottom right, connected to the circuit lines. Large dark blue geometric shapes, resembling stylized triangles or chevrons, are positioned in the top right and bottom left corners.

ELECTRICITY BILL GENERATOR

A web-based application for
calculating electricity bills

PROJECT OVERVIEW

The Electricity Bill Generator is a simple yet powerful web application designed to calculate electricity bills based on unit consumption. It implements a slab-based tariff system that accurately computes charges across different consumption tiers, providing users with a detailed bill breakdown.

This project demonstrates fundamental frontend development skills through practical implementation using HTML for structure, CSS for styling, and JavaScript for logic and interactivity. It showcases conditional logic, input validation, and dynamic content rendering in a real-world context.

KEY FEATURES

- Calculates electricity bill using slab-based pricing system
- Displays detailed bill breakdown with final amount
- Applies 5% surcharge on total bill automatically
- Validates user input for accuracy and error prevention
- Simple, clean user interface for easy navigation
- Reset functionality to clear inputs and results instantly

TECH STACK

HTML provides the structural foundation for the user interface, defining input fields, buttons, and result display areas. CSS handles all visual styling, creating a clean and professional appearance with responsive design elements.

JavaScript powers the core logic and interactivity, handling input validation, slab-based calculations, and dynamic content rendering. The application is browser-compatible (Chrome, Edge, Firefox) and developed using VS Code.

PROJECT STRUCTURE

- 1.** index.html (main interface), style.css (styles), script.js (bill calculation logic) – Clean separation of structure, presentation, and behavior.
- 2.** README.md (project documentation) – Modular separation of concerns ensures maintainability and scalability.

TARIFF DETAILS

- 0–50 units: ₹1.5 per unit
- 51–150 units: ₹2 per unit
- 151–250 units: ₹3 per unit
- Above 250 units: ₹4 per unit

Additionally, a 5% surcharge is applied to the total calculated bill.

HOW IT WORKS

- 1.** User inputs consumer name and units consumed, clicks "Generate Bill" button, which triggers JavaScript validation to ensure accurate input before processing.
- 2.** Bill is calculated using slab-based conditions, 5% surcharge is added to the total, detailed breakdown is displayed dynamically, and Reset button clears all inputs and results.

SYSTEM REQUIREMENTS

Modern Web Browser: Chrome, Edge, or Firefox required to run the application. The app uses standard HTML5, CSS3, and JavaScript features supported by all major browsers.

Visual Studio Code (Optional): Recommended code editor for development and editing. Any text editor can be used, but VS Code offers helpful extensions for HTML, CSS, and JavaScript development.

CONCLUSION

This Electricity Bill Generator project demonstrates how basic web technologies can solve real-world problems effectively. Through HTML, CSS, and JavaScript, we implemented conditional logic for slab-based tariff calculations, input validation to ensure data accuracy, and dynamic content rendering to display detailed bill breakdowns. The project reinforces fundamental frontend development concepts including DOM manipulation, event handling, and modular code organization. These skills form the foundation for building more complex web applications.

Thank you for your attention. Questions or live demo welcome!

Electricity Bill Generator

THANK YOU!

Questions or Demo Welcome